

Development and validation of an algorithm to establish a total antioxidant capacity database of the US diet.

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Estimation of total antioxidant intake is the first step to investigate the protective effects of antioxidants on oxidative stress-mediated disease. The present study was designed to develop an algorithm to estimate total antioxidant capacity (TAC) of the US diet. TAC of individual antioxidants and 50 popular antioxidant-rich food items in the US diet were determined by 2,2'-azino-bis-3-ethylbenzthiazoline-6-sulphonic acid (ABTS) assay and the 1,1-diphenyl-2-picrylhydrazyl (DPPH) assay. Theoretical TAC of foods was calculated as the sum of individual antioxidant capacities of compounds. The top 10 TAC food items in the US diet according to standard serving size were blueberry > plum > green tea > strawberry > green tea (decaffeinated) > red wine > grape juice > black tea > cherry > grape. Major contributors to TAC were the total phenolic content ($r = 0.952$, $P < 0.001$) and flavonoid content ($r = 0.827$, $P < 0.001$) of 50 foods. Theoretical TAC was positively correlated to experimental TAC of 50 foods determined by the ABTS assay ($r = 0.833$, $P < 0.001$) and the DPPH assay ($r = 0.696$, $P < 0.001$), and to TAC from the USDA database for the oxygen radical absorbance capacity ($r = 0.484$, $P = 0.001$, $n = 44$). The TAC database of the US diet has been established and validated. In future studies, TAC of the US diet can be linked to biomarkers of chronic disease.